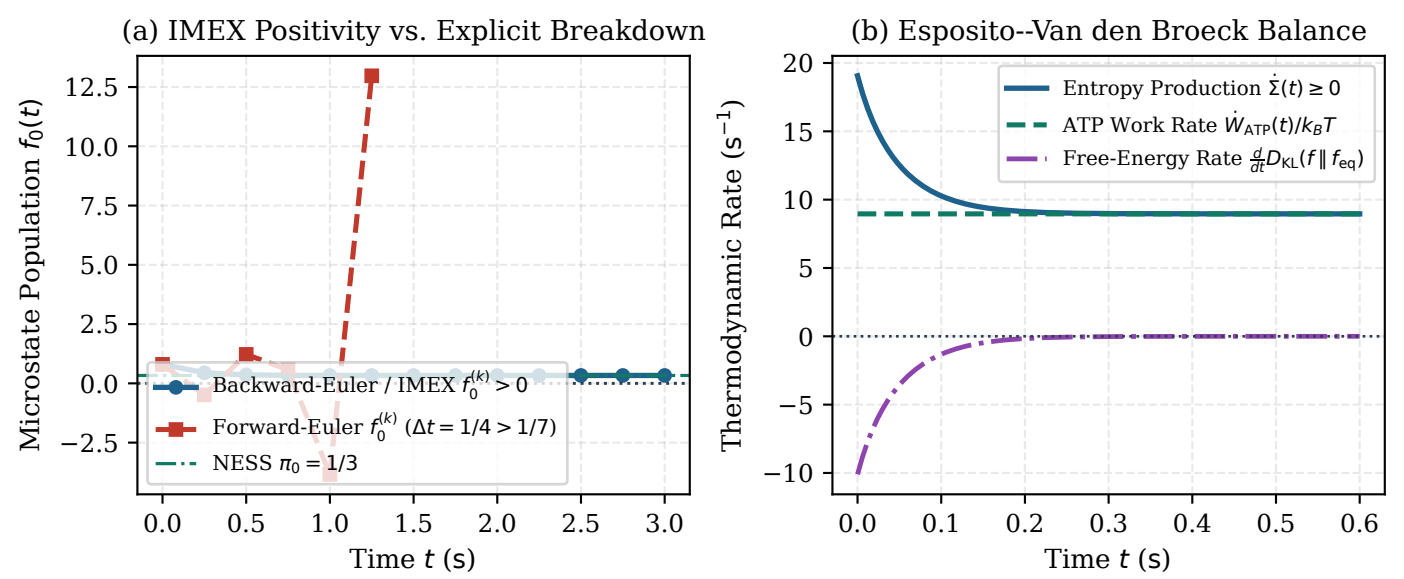
(a) IMEX Positivity vs. Explicit Breakdown
Microstate Population $f_0(t)$
Time t (s)
Backward-Euler / IMEX $f_0^{(k)} > 0$
Forward-Euler $f_0^{(k)}$ ($\Delta t = 1/4 > 1/7$)
NESS $\pi_0 = 1/3$
(b) Esposito--Van den Broeck Balance
Thermodynamic Rate (s^{-1})
Time t (s)
Entropy Production $\dot{\Sigma}(t) \geq 0$
ATP Work Rate $\dot{W}_{\mathrm{ATP}}(t)/k_B T$
Free-Energy Rate $\frac{d}{dt} D_{\mathrm{KL}}(f \| f_{\mathrm{eq}})$